EXACTNESS OF THE SKYSCRAPER SHEAF FUNCTOR P_* AND THE FUNCTORIAL SPLITTING P^{-1}P_*A = A \OPLUS I(A)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one same-chapter lemma and two Sites lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Adjunction unit and counit between p_* and p^{-1} for a point. Stacks Project, Sites, Lemma sites-lemma-stalk-skyscraper. Supplies the functorial maps A \to p^{-1}p_*A \to A whose composition is the identity, from which the splitting is built.
- Exactness of the pushforward along the localization morphism. Stacks Project, Modules on Sites, Lemma lemma-exactness-pushforward-pullback. Supplies left exactness of p_* : Ab \to Ab(C).
- Skyscraper sheaf functors are exact. Stacks Project, Sites, Lemma sites-lemma-skyscraper-functor-exact. Supplies that p_* transforms surjections into surjections.

Source reference. Original reference 05UX: sources/stacks-chow/tags.txt, source lines 7531–7531. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03DC: sources/stacks-chow/tags.txt, source lines 4415–4415. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05UY: sources/stacks-chow/tags.txt, source lines 7532–7532. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let p be a point of $\mathcal{C}$ or of its associated topos.*

(1) *The functor $p_* : Ab \to Ab(\mathcal{C})$, $A \mapsto p_*A$ is exact.*
(2) *There is a functorial direct sum decomposition*

$$p^{-1}p_*A = A \oplus I(A)$$

for $A \in \mathrm{Ob}(Ab)$.

Proof. By Sites, Lemma 05UX there are functorial maps $A \to p^{-1}p_*A \to A$ whose composition equals id_A. Hence a functorial direct sum decomposition as in (2) with $I(A)$ the kernel of the adjunction map $p^{-1}p_*A \to A$. The functor p_* is left exact by Lemma 03DC. The functor p_* transforms surjections into surjections by Sites, Lemma 05UY. Hence (1) holds. □

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